

Java & Spring Interview Mastery — Complete Study Plan

Section 1: Agile & Software Development Methodology

- Agile fundamentals, Scrum, Sprints, Iterations
- Stories, Story Points, Burn-down Charts
- Stand-up Meetings
- Agile vs Waterfall
- **Questions covered: 1-11**

Section 2: SOLID Principles & Design Patterns

- SOLID Principles (SRP, OCP, LSP, ISP, DIP)
- Design Principles overview
- Design Patterns (Creational, Structural, Behavioral)
- Facade Pattern (Logger/SLF4J context)
- **Questions covered: 12-15, 176**

Section 3: Java Fundamentals & Architecture

- Java features, history, ownership
- JDK vs JRE vs JVM
- Memory management, ClassLoader, Bytecode, JIT
- JAR files — creation, extraction, purpose
- **Questions covered: 16-29, 41**

Section 4: Data Types, Strings & String Internals

- Primitive & reference data types
- String, StringBuffer, StringBuilder
- String Pool, immutability, thread safety
- Scenarios for choosing between them
- **Questions covered: 30-40**

Section 5: OOP Core — Classes, Objects, Constructors

- Constructors, Copy Constructors
- Object class methods (clone, hashCode, equals, toString, etc.)

- Cloneable interface, deep vs shallow copy
- Static keyword, static blocks
- **Questions covered: 42-53, 172**

Section 6: OOP — Inheritance, Abstraction, Polymorphism

- Abstract classes vs Interfaces
- Access modifiers (public, protected, default, private)
- Inheritance types, Diamond Problem
- IS-A vs HAS-A, Aggregation, Containment
- Method Overloading vs Overriding
- `this` and `super` keywords
- Packages, Encapsulation (getters/setters)
- **Questions covered: 54-76, 179a, 179b, 179c**

Section 7: Exception Handling

- Checked vs Unchecked Exceptions
- Exception hierarchy (Throwable → Error/Exception)
- throw vs throws, custom exceptions
- try-catch-finally, multiple catch blocks
- **Questions covered: 77-88, 179**

Section 8: Generics

- Generic classes, methods, bounded types
- Wildcards (?, extends, super)
- Type erasure
- **Questions covered: 89-91**

Section 9: Collections Framework (Deep Dive)

- Collection hierarchy (Iterable → Collection → List/Set/Queue)
- ArrayList, LinkedList, Vector
- HashSet, TreeSet
- HashMap, TreeMap, LinkedHashMap
- Iterator vs ListIterator
- Comparable vs Comparator

- Sorting objects by attributes
- Thread-safe collections (ConcurrentHashMap vs SynchronizedMap)
- **Questions covered: 92-118, 130, 171**

Section 10: Multithreading & Concurrency

- Thread creation (Thread class, Runnable, Callable)
- Thread lifecycle & states
- wait vs sleep, join, synchronization
- Synchronized blocks, Executor Service, Thread Pool
- **Questions covered: 38-39, 106-108, 119-129, 171 (sync block)**

Section 11: Java 8+ Features

- Lambda expressions, Functional Interfaces
- Consumer, Supplier, Function, Predicate
- Method referencing
- Streams & Parallel Streams
- Optional class
- LocalDate, LocalTime vs Date
- Nashorn JS engine
- **Questions covered: 131-143, 167-168, 179d**

Section 12: Modern Java (Sealed Classes, Records, DTOs)

- Sealed classes
- Records and their use cases
- DTO vs DAO pattern
- **Questions covered: 144-149**

Section 13: Java Reflection & Runtime

- Reflection API, loading classes at runtime
- `getClass()`, `Class.forName()`
- **Questions covered: 109, 169-170**

Section 14: Serialization, Cloning & Annotations

- Serializable & Cloneable interfaces

- @Override annotation behavior
- **Questions covered: 173-174**

Section 15: Logging

- Logger, SLF4J, Facade pattern
- Log levels, implementations (Logback, Log4j)
- **Questions covered: 175-177**

Section 16: Maven

- POM.xml structure (groupId, artifactId, version)
- Dependency management & scopes
- Maven lifecycle commands (clean, validate, compile, test, install)
- Maven plugins, Maven Central
- **Questions covered: 150-166**

Section 17: Git

- 10 essential Git commands with explanations
- **Questions covered: 178**

Section 18: Reactive Programming

- Mono & Flux
- Asynchronous programming
- Project Reactor
- Backpressure
- **Questions covered: 180-183**

Section 19: Spring Framework Core

- Spring Container, Dependency Injection
- BeanFactory vs ApplicationContext
- Bean scopes, lifecycle
- Constructor vs Setter injection
- Eager vs Lazy loading
- Annotations: @Component, @Service, @Repository, @Controller

- @Bean vs @Component, @Qualifier, @Value, @PropertySource
- Prototype scope

Section 20: Spring Boot

- Spring Boot vs Spring Framework
- @SpringBootApplication, Auto-configuration
- Starters, REST API creation
- application.properties vs application.yml
- Global exception handling
- Spring Boot Actuator

Section 21: Spring MVC

- DispatcherServlet, MVC architecture
- @Controller vs @RestController
- @RequestMapping, @PathVariable, @RequestParam
- Form handling, View resolution
- HandlerInterceptor, Exception handling

Section 22: Testing (JUnit & Mockito)

- JUnit basics
- @Before, @BeforeEach, @After, @AfterEach
- Testing Spring beans
- @MockBean vs @Mock

How we'll work: I'll teach each section in detail with explanations, code examples, diagrams, analogies, and interview tips. Say "next" anytime to move to the next section.